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Expectiminimax Depth Optimization

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Games per configuration: 10

Roll depth range: 1-5

Stop depth range: 1-6

Seed: 42

Time budget: 50 ms per move (iterative deepening)

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|         |                                     |                 |                                        |
|---------|-------------------------------------|-----------------|----------------------------------------|
| [1/30]  | Testing rollDepth=1, stopDepth=1... | Win rate: 60,0% | (avg moves: 57,0, avg time: 2864,7 ms) |
| [2/30]  | Testing rollDepth=1, stopDepth=2... | Win rate: 60,0% | (avg moves: 60,2, avg time: 3016,2 ms) |
| [3/30]  | Testing rollDepth=1, stopDepth=3... | Win rate: 50,0% | (avg moves: 61,7, avg time: 3092,1 ms) |
| [4/30]  | Testing rollDepth=1, stopDepth=4... | Win rate: 30,0% | (avg moves: 66,2, avg time: 3317,4 ms) |
| [5/30]  | Testing rollDepth=1, stopDepth=5... | Win rate: 70,0% | (avg moves: 66,7, avg time: 3341,4 ms) |
| [6/30]  | Testing rollDepth=1, stopDepth=6... | Win rate: 80,0% | (avg moves: 64,0, avg time: 3209,6 ms) |
| [7/30]  | Testing rollDepth=2, stopDepth=1... | Win rate: 60,0% | (avg moves: 60,0, avg time: 3006,2 ms) |
| [8/30]  | Testing rollDepth=2, stopDepth=2... | Win rate: 70,0% | (avg moves: 66,4, avg time: 3327,4 ms) |
| [9/30]  | Testing rollDepth=2, stopDepth=3... | Win rate: 50,0% | (avg moves: 64,6, avg time: 3236,9 ms) |
| [10/30] | Testing rollDepth=2, stopDepth=4... | Win rate: 70,0% | (avg moves: 60,3, avg time: 3022,2 ms) |
| [11/30] | Testing rollDepth=2, stopDepth=5... | Win rate: 60,0% | (avg moves: 64,9, avg time: 3251,5 ms) |
| [12/30] | Testing rollDepth=2, stopDepth=6... | Win rate: 30,0% | (avg moves: 66,5, avg time: 3331,4 ms) |
| [13/30] | Testing rollDepth=3, stopDepth=1... | Win rate: 30,0% | (avg moves: 61,0, avg time: 3057,0 ms) |
| [14/30] | Testing rollDepth=3, stopDepth=2... | Win rate: 40,0% | (avg moves: 63,2, avg time: 3166,3 ms) |
| [15/30] | Testing rollDepth=3, stopDepth=3... | Win rate: 50,0% | (avg moves: 55,3, avg time: 2773,5 ms) |
| [16/30] | Testing rollDepth=3, stopDepth=4... | Win rate: 80,0% | (avg moves: 56,4, avg time: 2825,2 ms) |
| [17/30] | Testing rollDepth=3, stopDepth=5... | Win rate: 40,0% | (avg moves: 58,0, avg time: 2906,8 ms) |
| [18/30] | Testing rollDepth=3, stopDepth=6... | Win rate: 50,0% | (avg moves: 71,4, avg time: 3577,2 ms) |
| [19/30] | Testing rollDepth=4, stopDepth=1... | Win rate: 70,0% | (avg moves: 61,5, avg time: 3082,0 ms) |
| [20/30] | Testing rollDepth=4, stopDepth=2... | Win rate: 40,0% | (avg moves: 66,0, avg time: 3307,1 ms) |
| [21/30] | Testing rollDepth=4, stopDepth=3... | Win rate: 40,0% | (avg moves: 58,0, avg time: 2906,9 ms) |
| [22/30] | Testing rollDepth=4, stopDepth=4... | Win rate: 70,0% | (avg moves: 61,1, avg time: 3060,8 ms) |
| [23/30] | Testing rollDepth=4, stopDepth=5... | Win rate: 80,0% | (avg moves: 70,2, avg time: 3521,5 ms) |
| [24/30] | Testing rollDepth=4, stopDepth=6... | Win rate: 70,0% | (avg moves: 62,0, avg time: 3105,9 ms) |
| [25/30] | Testing rollDepth=5, stopDepth=1... | Win rate: 70,0% | (avg moves: 64,2, avg time: 3222,9 ms) |
| [26/30] | Testing rollDepth=5, stopDepth=2... | Win rate: 40,0% | (avg moves: 60,1, avg time: 3011,3 ms) |
| [27/30] | Testing rollDepth=5, stopDepth=3... | Win rate: 50,0% | (avg moves: 70,5, avg time: 3531,7 ms) |
| [28/30] | Testing rollDepth=5, stopDepth=4... | Win rate: 40,0% | (avg moves: 67,0, avg time: 3359,7 ms) |
| [29/30] | Testing rollDepth=5, stopDepth=5... | Win rate: 70,0% | (avg moves: 60,3, avg time: 3021,0 ms) |
| [30/30] | Testing rollDepth=5, stopDepth=6... | Win rate: 80,0% | (avg moves: 61,5, avg time: 3081,6 ms) |

# OPTIMIZATION RESULTS

| Roll | Stop | Win Rate | Avg Moves | Avg Time |    |
|------|------|----------|-----------|----------|----|
| 1    | 6    | 80,0     | % 64,0    | 3209,6   | ms |
| 3    | 4    | 80,0     | % 56,4    | 2825,2   | ms |
| 4    | 5    | 80,0     | % 70,2    | 3521,5   | ms |
| 5    | 6    | 80,0     | % 61,5    | 3081,6   | ms |
| 1    | 5    | 70,0     | % 66,7    | 3341,4   | ms |
| 2    | 2    | 70,0     | % 66,4    | 3327,4   | ms |
| 2    | 4    | 70,0     | % 60,3    | 3022,2   | ms |
| 4    | 1    | 70,0     | % 61,5    | 3082,0   | ms |
| 4    | 4    | 70,0     | % 61,1    | 3060,8   | ms |
| 4    | 6    | 70,0     | % 62,0    | 3105,9   | ms |
| 5    | 1    | 70,0     | % 64,2    | 3222,9   | ms |
| 5    | 5    | 70,0     | % 60,3    | 3021,0   | ms |
| 1    | 1    | 60,0     | % 57,0    | 2864,7   | ms |
| 1    | 2    | 60,0     | % 60,2    | 3016,2   | ms |
| 2    | 1    | 60,0     | % 60,0    | 3006,2   | ms |
| 2    | 5    | 60,0     | % 64,9    | 3251,5   | ms |
| 1    | 3    | 50,0     | % 61,7    | 3092,1   | ms |
| 2    | 3    | 50,0     | % 64,6    | 3236,9   | ms |
| 3    | 3    | 50,0     | % 55,3    | 2773,5   | ms |
| 3    | 6    | 50,0     | % 71,4    | 3577,2   | ms |
| 5    | 3    | 50,0     | % 70,5    | 3531,7   | ms |
| 3    | 2    | 40,0     | % 63,2    | 3166,3   | ms |
| 3    | 5    | 40,0     | % 58,0    | 2906,8   | ms |
| 4    | 2    | 40,0     | % 66,0    | 3307,1   | ms |
| 4    | 3    | 40,0     | % 58,0    | 2906,9   | ms |
| 5    | 2    | 40,0     | % 60,1    | 3011,3   | ms |
| 5    | 4    | 40,0     | % 67,0    | 3359,7   | ms |
| 1    | 4    | 30,0     | % 66,2    | 3317,4   | ms |
| 2    | 6    | 30,0     | % 66,5    | 3331,4   | ms |
| 3    | 1    | 30,0     | % 61,0    | 3057,0   | ms |

## --- Top 3 Configurations ---

1. rollDepth=1, stopDepth=6: 80,0% win rate
2. rollDepth=3, stopDepth=4: 80,0% win rate
3. rollDepth=4, stopDepth=5: 80,0% win rate

## --- Best Balanced (performance/time) ---

rollDepth=1, stopDepth=6: 80,0% win rate, 3209,6 ms avg